

Sanjeevraj Bommannan M

Electronics & Communication Engineering Undergraduate



Electronics and Communication Engineering undergraduate with strong fundamentals in VLSI Design, Digital Electronics, Verilog/SystemVerilog, and FPGA development. Seeking to leverage my technical skills and project experience in a VLSI Design or Design Verification role while contributing to the development of reliable and high-performance semiconductor solutions.

+91 7708044151 / [✉ sanjeevrajbm2006@gmail.com](mailto:sanjeevrajbm2006@gmail.com) / [in](https://www.linkedin.com/in/bomman) linkedin.com/in/bomman / github.com/srb6
5/3, Chamundi Street, Gugai, Salem – 636006

PROJECTS

Smart Energy Monitoring System

- Designed a real-time energy monitoring and automation system using ESP32, ACS712 current sensors and voltage sensors with LCD and cloud dashboard integration.
- Integrated IoT automation and security features including relay control, energy analytics, EB billing estimation and remote alerts.

Smart Shopping Trolley

- Developed a Smart Shopping Trolley system using load cell sensor, and Firebase for real-time automated billing with anti-theft protection.
- Designed a mobile application with barcode scanning and digital payment integration to improve checkout efficiency and enhance payment methods in retail environments.

Multi-Wavelength Optical Microplastic Detection System

- Developed an embedded optical sensing system using multi-wavelength LEDs (UV, Visible, IR), and photodiodes to detect microplastics in water using light scattering and absorption principles.
- Implemented real-time signal processing and TinyML-based analysis to estimate microplastic concentration and enable portable water quality monitoring.

ACHIEVEMENTS

- Ideathon – Second Place
- Techno Fest (PoC) – Top 3
- 30 Hours Hackathon – Second Place

EDUCATION

B.E Electronics and Communication Engineering Kongu Engineering College

Perundurai | 2024 – Present
CGPA: 8.82

Higher Secondary Education
Sri Sarada Balamandir Boys Matric. Hr. Sec. School
Salem | 2010 – 2024
Percentage: 83.83%

TECHNICAL SKILL

Programming: C, C++, Python, Verilog,
System Verilog, VHDL
Microcontroller: 8051, ESP32, Arduino
Operating System: Ubuntu Linux

TOOLS

Proteus, Altium, Cadence (Genus, Pspice),
ModelSim, QuestaSim, MATLAB, VS Code, LTSpice

AREAS OF INTEREST

VLSI Design
Embedded Systems

SOFT SKILLS

Problem Solving
Team Collaboration
Time Management

LANGUAGES

Kannada (Native)
Tamil (Conversational)
English (Professional)
Hindi (Beginner)